CampusPlus — GRASP-Compliant Class Diagram
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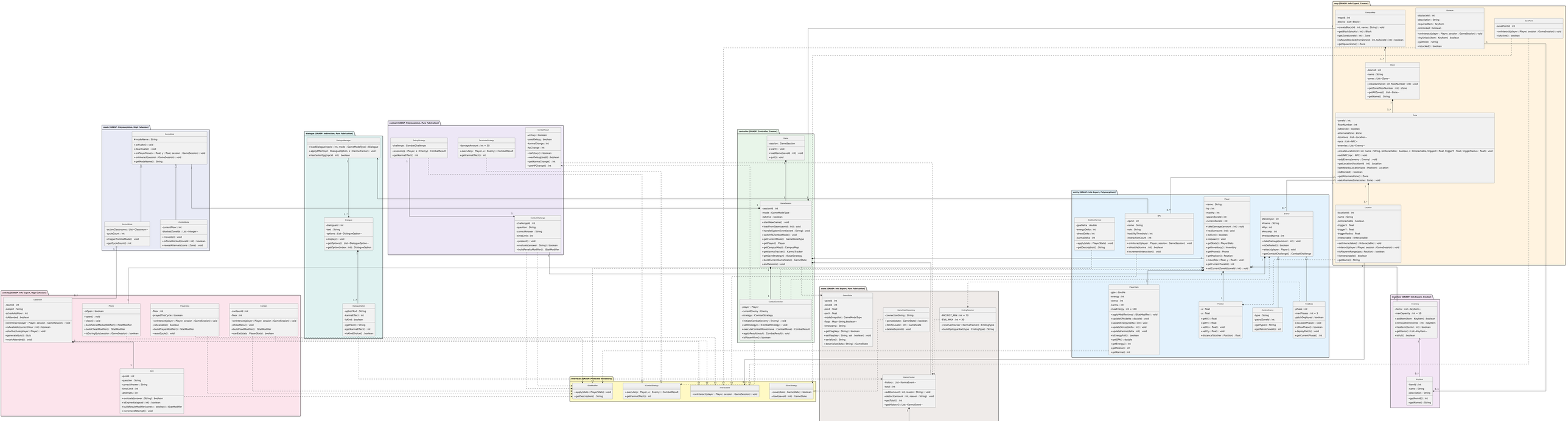
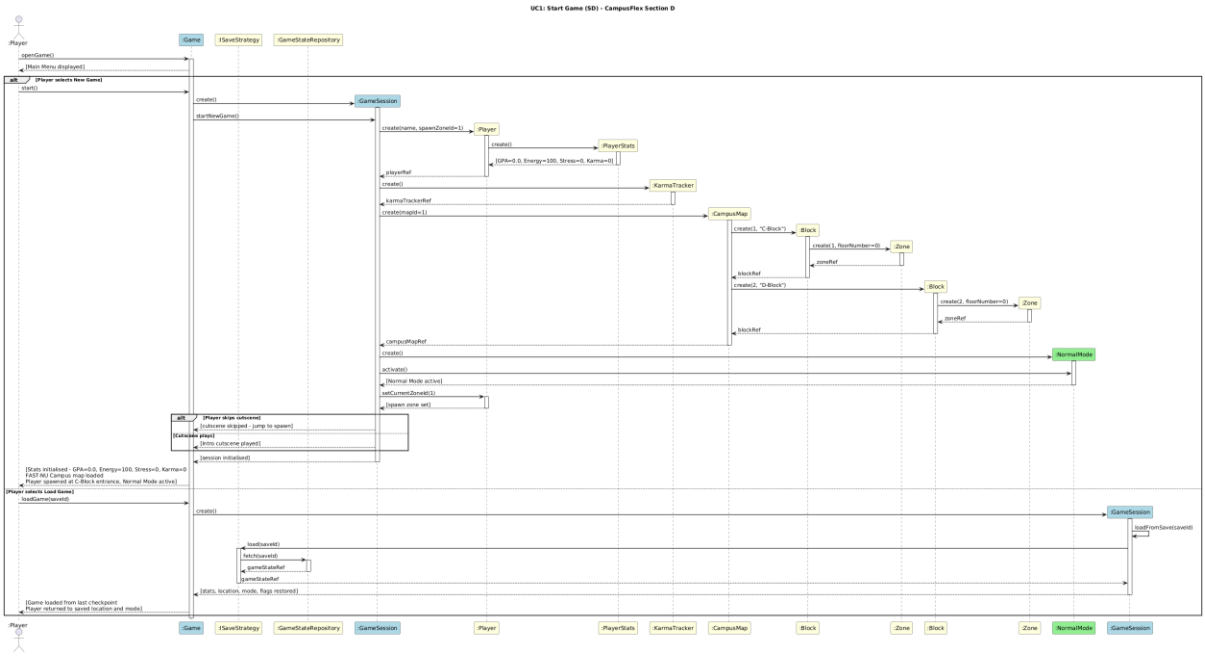
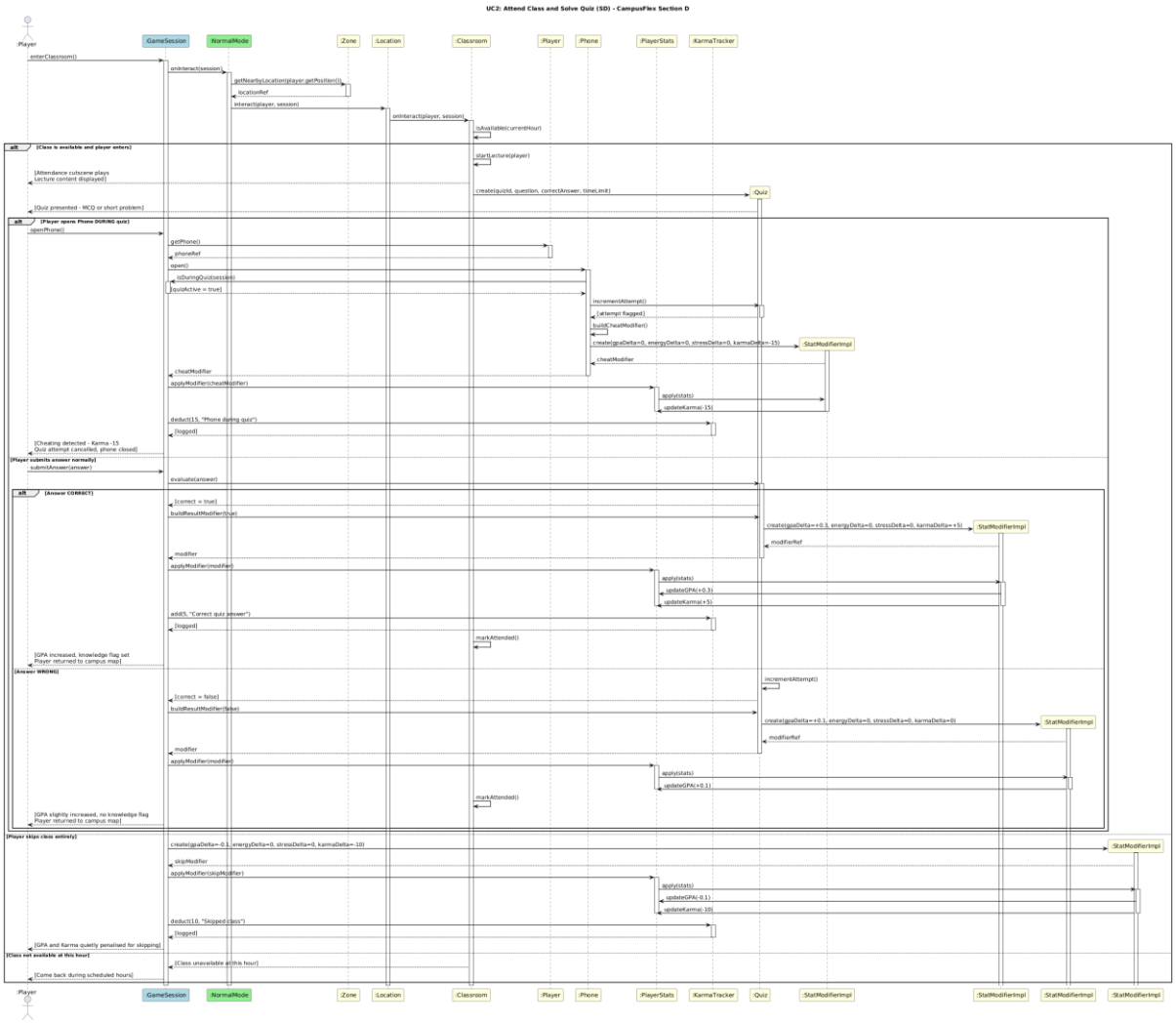


CampusFlex – State Diagrams

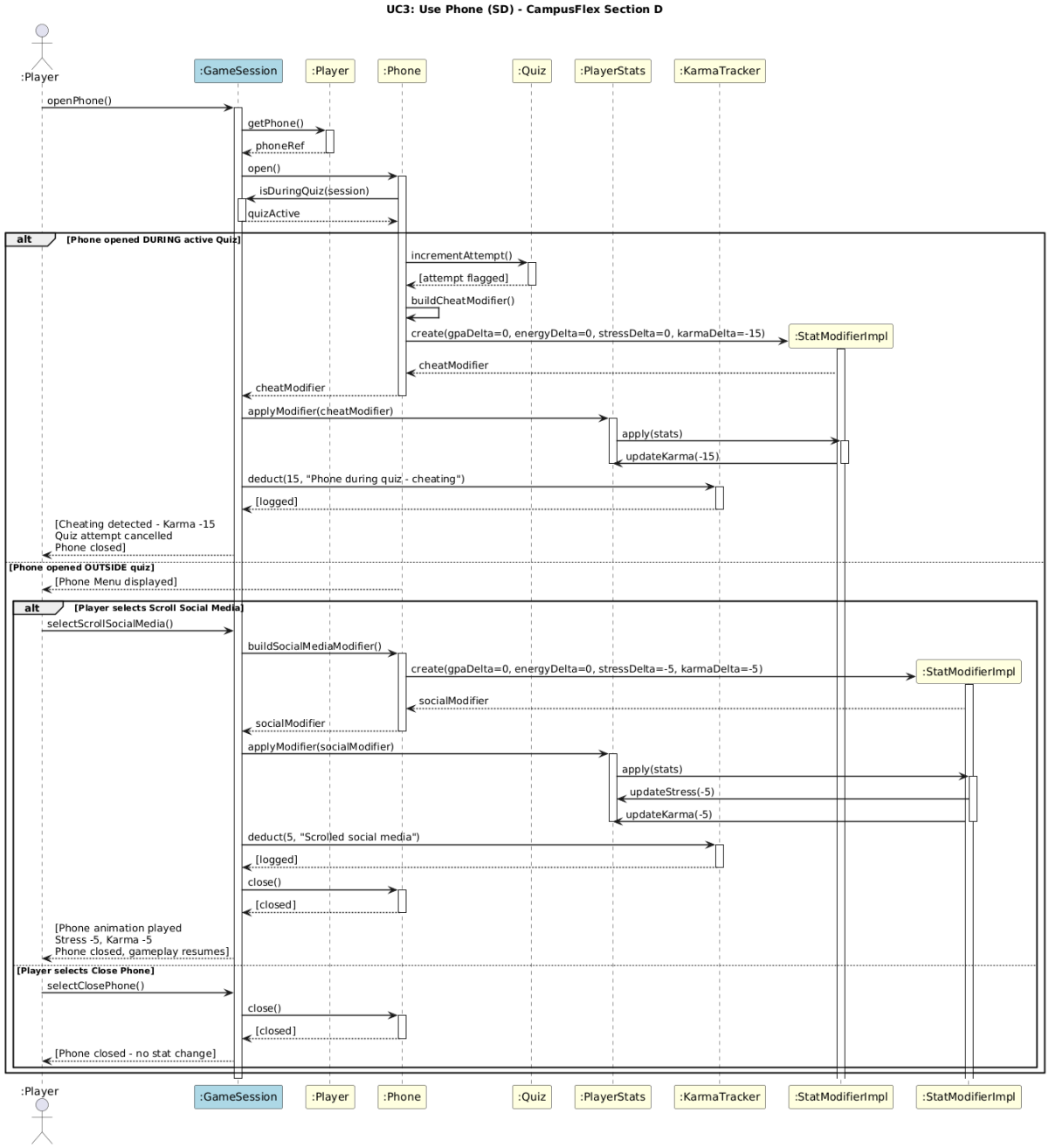
Use Case 1



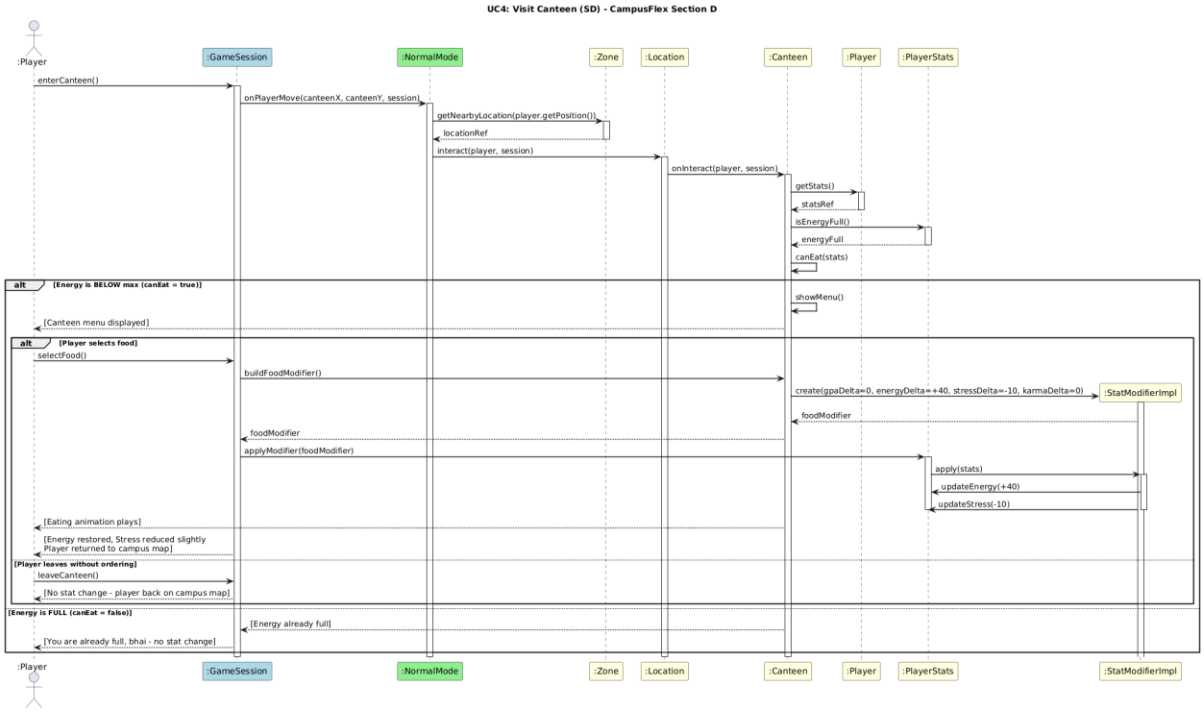
Use Case 2


UC2: Attend Class and Solve Quiz (SD) - CampusFlex Section D
:Player
:GameSession
:NormalMode
:Zone
:Location
:Classroom
:Player
:Phone
:PlayerStats
:KarmaTracker
enterClassroom()
onInteract(session)
getNearbyLocation(player.getPosition())
locationRef
interact(player, session)
onInteract(player, session)
isAvailable(currentHour)
alt
[Class is available and player enters]
startLecture(player)
[Attendance cutscene plays
Lecture content displayed]
create(quizId, question, correctAnswer, timeLimit)
:Quiz
[Quiz presented - MCQ or short problem]
alt
[Player opens Phone DURING quiz]
openPhone()
getPhone()
phoneRef
open()
isDuringQuiz(session)
[quizActive = true]
incrementAttempt()
[attempt flagged]
buildCheatModifier()
create(gpaDelta=0, energyDelta=0, stressDelta=0, karmaDelta=-15)
:StatModifierImpl
cheatModifier
cheatModifier
applyModifier(cheatModifier)
apply(stats)
updateKarma(-15)
deduct(15, "Phone during quiz")
[logged]
[Cheating detected - Karma -15
Quiz attempt cancelled, phone closed]
[Player submits answer normally]
submitAnswer(answer)
evaluate(answer)
alt
[Answer CORRECT]
[correct = true]
buildResultModifier(true)
create(gpaDelta=+0.3, energyDelta=0, stressDelta=0, karmaDelta=+5)
:StatModifierImpl
modifierRef
modifier
applyModifier(modifier)
apply(stats)
updateGPA(+0.3)
updateKarma(+5)
add(5, "Correct quiz answer")
[logged]
markAttended()
[GPA increased, knowledge flag set
Player returned to campus map]
[Answer WRONG]
incrementAttempt()
[correct = false]
buildResultModifier(false)
create(gpaDelta=+0.1, energyDelta=0, stressDelta=0, karmaDelta=0)
:StatModifierImpl
modifierRef
modifier
applyModifier(modifier)
apply(stats)
updateGPA(+0.1)
markAttended()
[GPA slightly increased, no knowledge flag
Player returned to campus map]
[Player skips class entirely]
create(gpaDelta=-0.1, energyDelta=0, stressDelta=0, karmaDelta=-10)
:StatModifierImpl
skipModifier
applyModifier(skipModifier)
apply(stats)
updateGPA(-0.1)
updateKarma(-10)
deduct(10, "Skipped class")
[logged]
[GPA and Karma quietly penalised for skipping]
[Class not available at this hour]
[Class unavailable at this hour]
[Come back during scheduled hours]
:Player
:GameSession
:NormalMode
:Zone
:Location
:Classroom
:Player
:Phone
:PlayerStats
:KarmaTracker
:Quiz
:StatModifierImpl
:StatModifierImpl
:StatModifierImpl
:StatModifierImpl

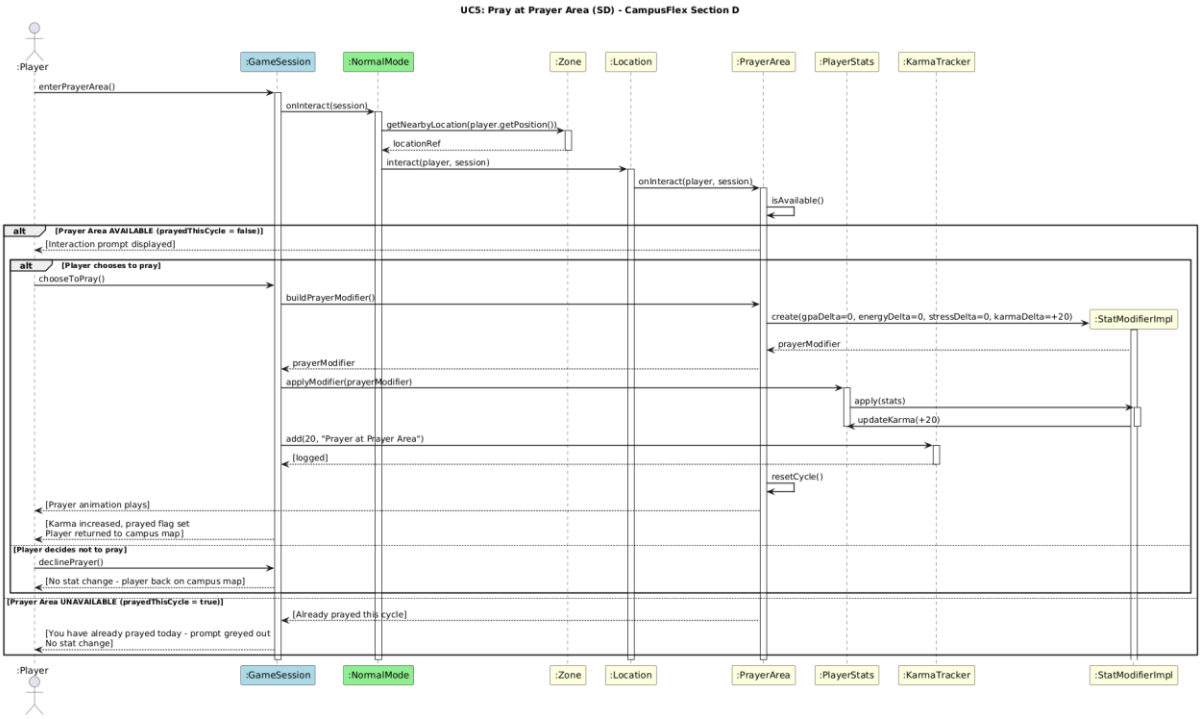
Use Case 3


UC3: Use Phone (SD) - CampusFlex Section D
:Player
:GameSession
:Player
:Phone
:Quiz
:PlayerStats
:KarmaTracker
openPhone()
getPhone()
phoneRef
open()
isDuringQuiz(session)
quizActive
alt
[Phone opened DURING active Quiz]
incrementAttempt()
[attempt flagged]
buildCheatModifier()
create(gpaDelta=0, energyDelta=0, stressDelta=0, karmaDelta=-15)
:StatModifierImpl
cheatModifier
cheatModifier
applyModifier(cheatModifier)
apply(stats)
updateKarma(-15)
deduct(15, "Phone during quiz - cheating")
[logged]
[Cheating detected - Karma -15
Quiz attempt cancelled
Phone closed]
[Phone opened OUTSIDE quiz]
[Phone Menu displayed]
alt
[Player selects Scroll Social Media]
selectScrollSocialMedia()
buildSocialMediaModifier()
create(gpaDelta=0, energyDelta=0, stressDelta=-5, karmaDelta=-5)
:StatModifierImpl
socialModifier
socialModifier
applyModifier(socialModifier)
apply(stats)
updateStress(-5)
updateKarma(-5)
deduct(5, "Scrolled social media")
[logged]
close()
[closed]
[Phone animation played
Stress -5, Karma -5
Phone closed, gameplay resumes]
[Player selects Close Phone]
selectClosePhone()
close()
[closed]
[Phone closed - no stat change]
:StatModifierImpl

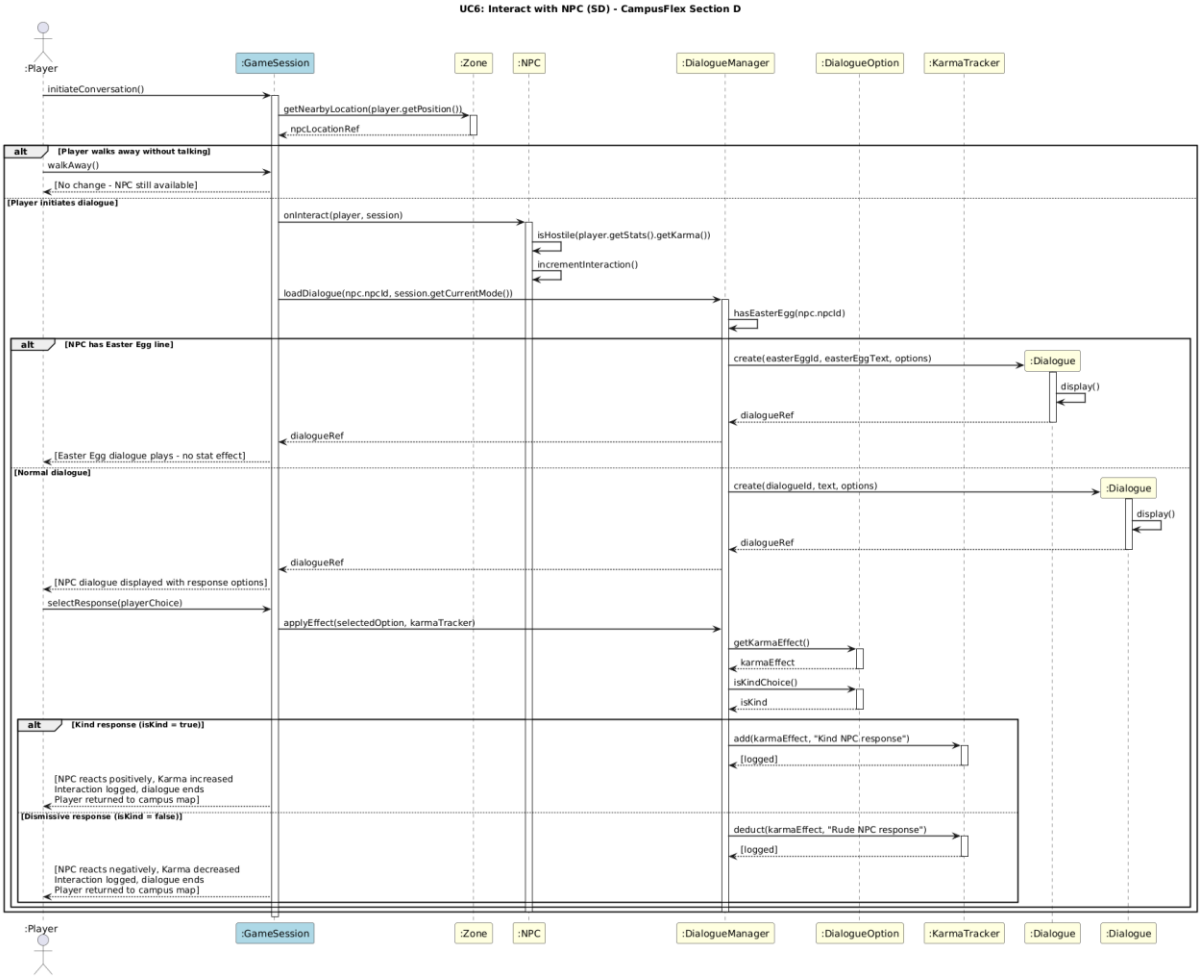
Use Case 4


UC4: Visit Canteen (SD) - CampusFlex Section D
:Player
:GameSession
:NormalMode
:Zone
:Location
:Canteen
:Player
:PlayerStats
:StatModifierImpl
enterCanteen()
onPlayerMove(canteenX, canteenY, session)
getNearbyLocation(player.getPosition())
locationRef
interact(player, session)
onInteract(player, session)
getStats()
statsRef
isEnergyFull()
energyFull
canEat(stats)
alt
[Energy is BELOW max (canEat = true)]
showMenu()
[Canteen menu displayed]
alt
[Player selects food]
selectFood()
buildFoodModifier()
create(gpaDelta=0, energyDelta=+40, stressDelta=-10, karmaDelta=0)
foodModifier
foodModifier
applyModifier(foodModifier)
apply(stats)
updateEnergy(+40)
updateStress(-10)
[Eating animation plays]
[Energy restored, Stress reduced slightly
Player returned to campus map]
[Player leaves without ordering]
leaveCanteen()
[No stat change - player back on campus map]
[Energy is FULL (canEat = false)]
[Energy already full]
[You are already full, bhai - no stat change]

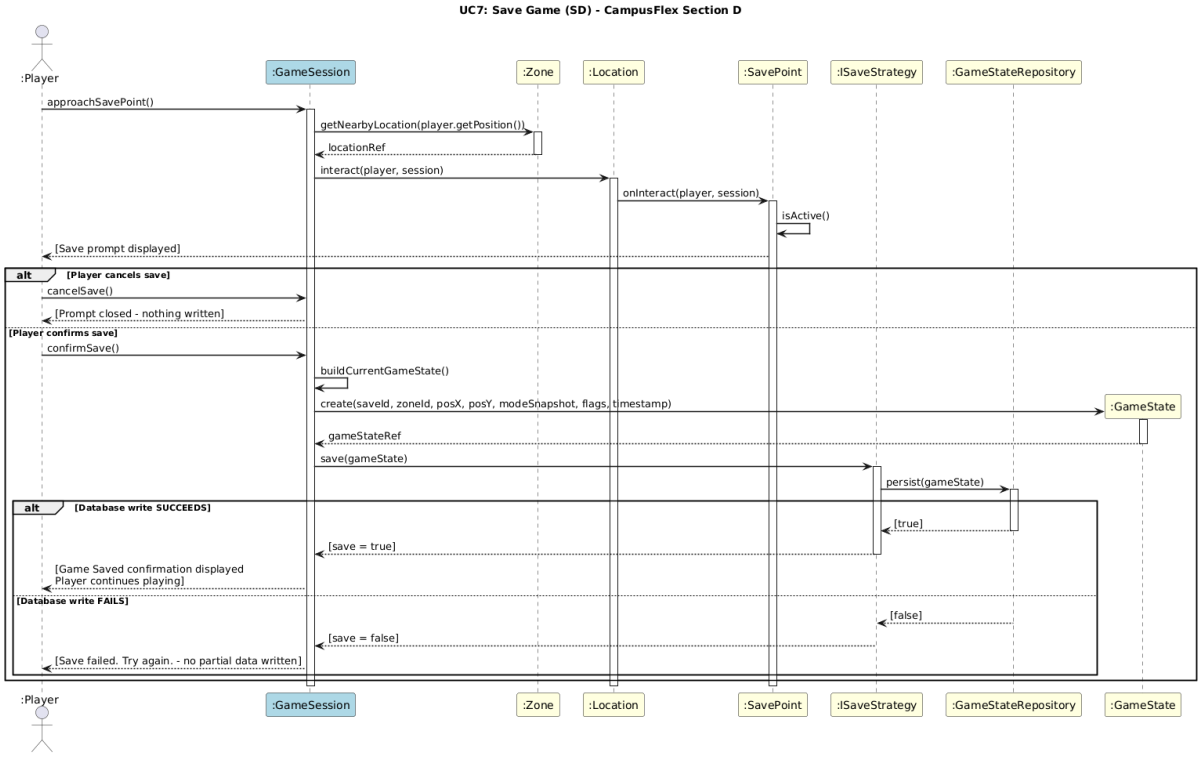
Use Case 5


UC5: Pray at Prayer Area (SD) - CampusFlex Section D
:Player
:GameSession
:NormalMode
:Zone
:Location
:PrayerArea
:PlayerStats
:KarmaTracker
:StatModifierImpl
enterPrayerArea()
onInteract(session)
getNearbyLocation(player.getPosition())
locationRef
interact(player, session)
onInteract(player, session)
isAvailable()
alt
[Prayer Area AVAILABLE (prayedThisCycle = false)]
[Interaction prompt displayed]
alt
[Player chooses to pray]
chooseToPray()
buildPrayerModifier()
create(gpaDelta=0, energyDelta=0, stressDelta=0, karmaDelta=+20)
prayerModifier
prayerModifier
applyModifier(prayerModifier)
apply(stats)
updateKarma(+20)
add(20, "Prayer at Prayer Area")
[logged]
resetCycle()
[Prayer animation plays]
[Karma increased, prayed flag set
Player returned to campus map]
[Player decides not to pray]
declinePrayer()
[No stat change - player back on campus map]
[Prayer Area UNAVAILABLE (prayedThisCycle = true)]
[Already prayed this cycle]
[You have already prayed today - prompt greyed out
No stat change]

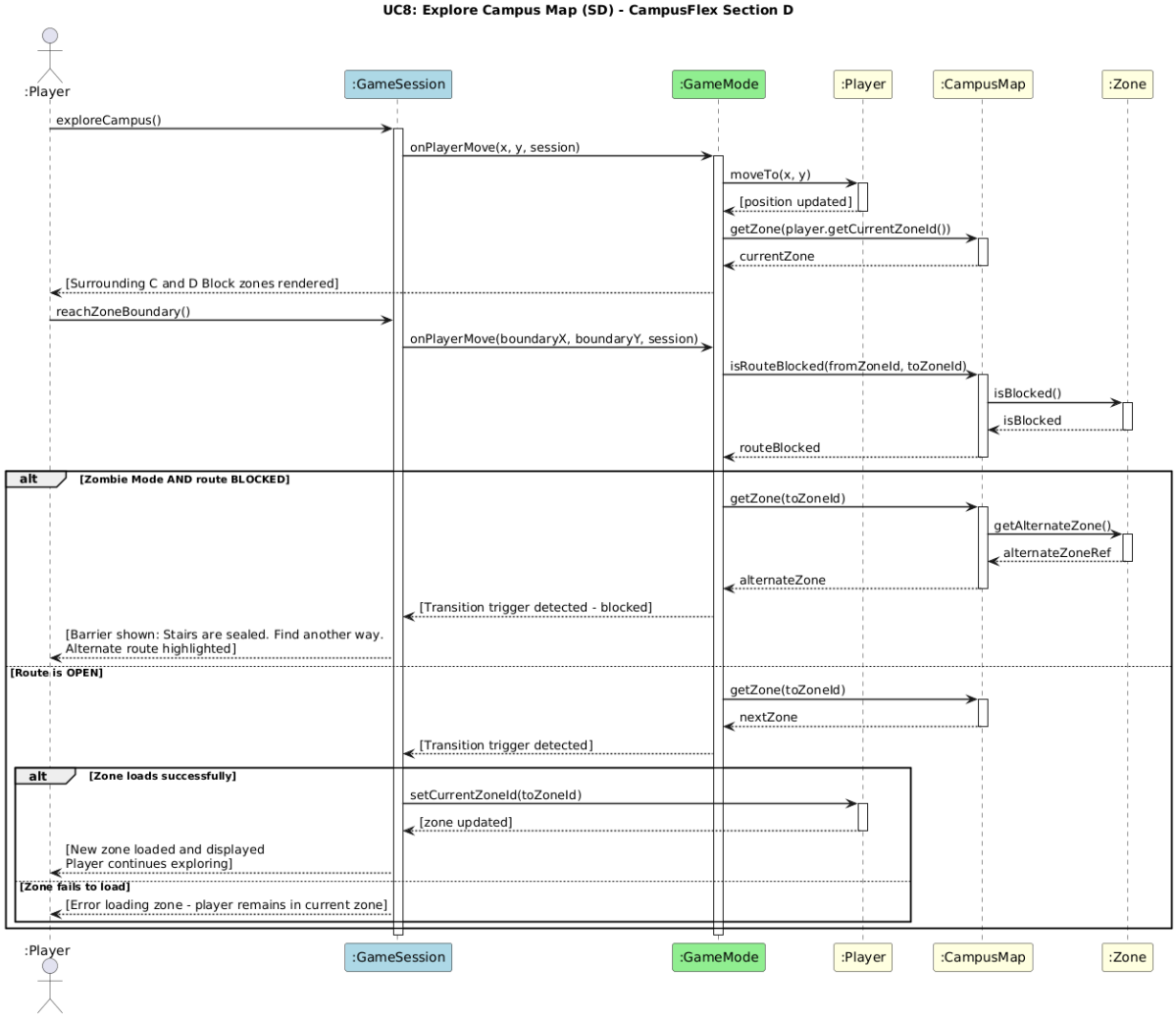
Use Case 6


UC6: Interact with NPC (SD) - CampusFlex Section D
:Player
:GameSession
:Zone
:NPC
:DialogueManager
:DialogueOption
:KarmaTracker
initiateConversation()
getNearbyLocation(player.getPosition())
npcLocationRef
alt
[Player walks away without talking]
walkAway()
[No change - NPC still available]
[Player initiates dialogue]
onInteract(player, session)
isHostile(player.getStats().getKarma())
incrementInteraction()
loadDialogue(npc.npcId, session.getCurrentMode())
hasEasterEgg(npc.npcId)
alt
[NPC has Easter Egg line]
create(easterEggId, easterEggText, options)
:Dialogue
display()
dialogueRef
dialogueRef
[Easter Egg dialogue plays - no stat effect]
[Normal dialogue]
create(dialogueId, text, options)
:Dialogue
display()
dialogueRef
dialogueRef
[NPC dialogue displayed with response options]
selectResponse(playerChoice)
applyEffect(selectedOption, karmaTracker)
getKarmaEffect()
karmaEffect
isKindChoice()
isKind
alt
[Kind response (isKind = true)]
add(karmaEffect, "Kind NPC response")
[logged]
[NPC reacts positively, Karma increased
Interaction logged, dialogue ends
Player returned to campus map]
[Dismissive response (isKind = false)]
deduct(karmaEffect, "Rude NPC response")
[logged]
[NPC reacts negatively, Karma decreased
Interaction logged, dialogue ends
Player returned to campus map]
:Player
:GameSession
:Zone
:NPC
:DialogueManager
:DialogueOption
:KarmaTracker
:Dialogue
:Dialogue

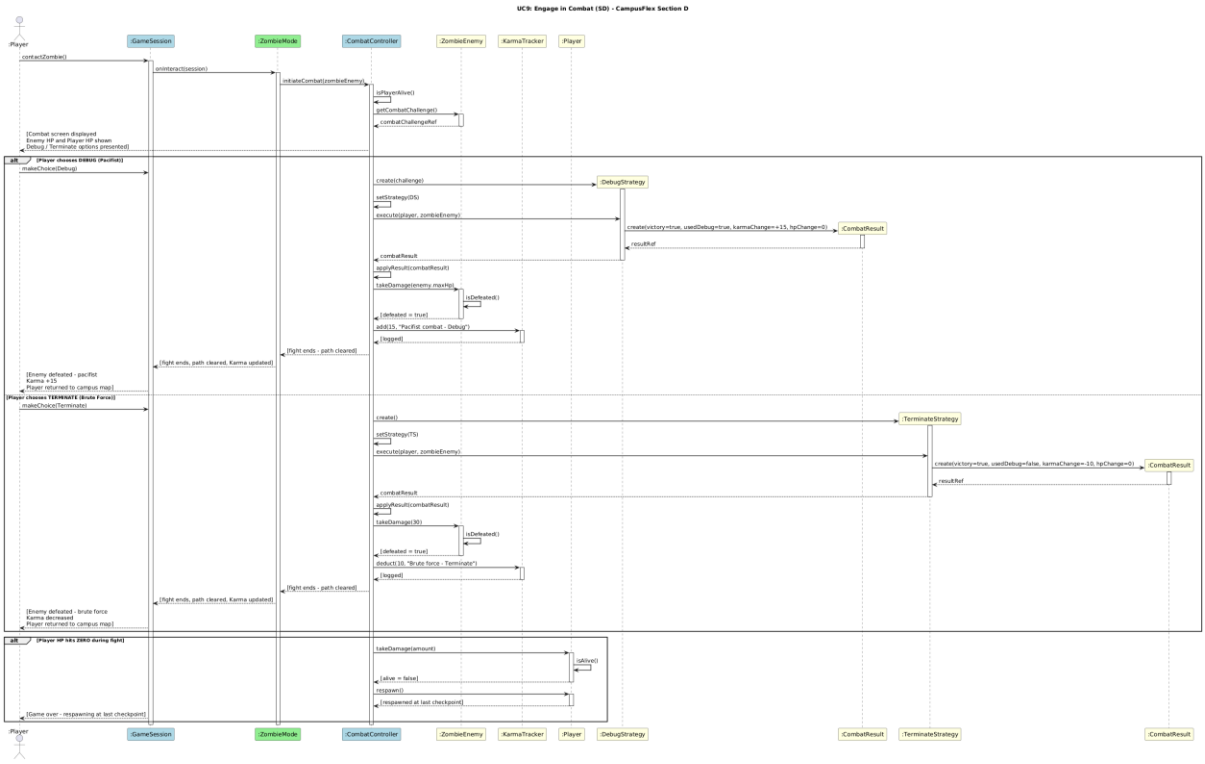
Use Case 7


UC7: Save Game (SD) - CampusFlex Section D
:Player
:GameSession
:Zone
:Location
:SavePoint
:ISaveStrategy
:GameStateRepository
approachSavePoint()
getNearbyLocation(player.getPosition())
locationRef
interact(player, session)
onInteract(player, session)
isActive()
[Save prompt displayed]
alt
[Player cancels save]
cancelSave()
[Prompt closed - nothing written]
[Player confirms save]
confirmSave()
buildCurrentGameState()
create(saveId, zoneId, posX, posY, modeSnapshot, flags, timestamp)
:GameState
gameStateRef
save(gameState)
persist(gameState)
alt
[Database write SUCCEEDS]
[true]
[save = true]
[Game Saved confirmation displayed
Player continues playing]
[Database write FAILS]
[false]
[save = false]
[Save failed. Try again. - no partial data written]

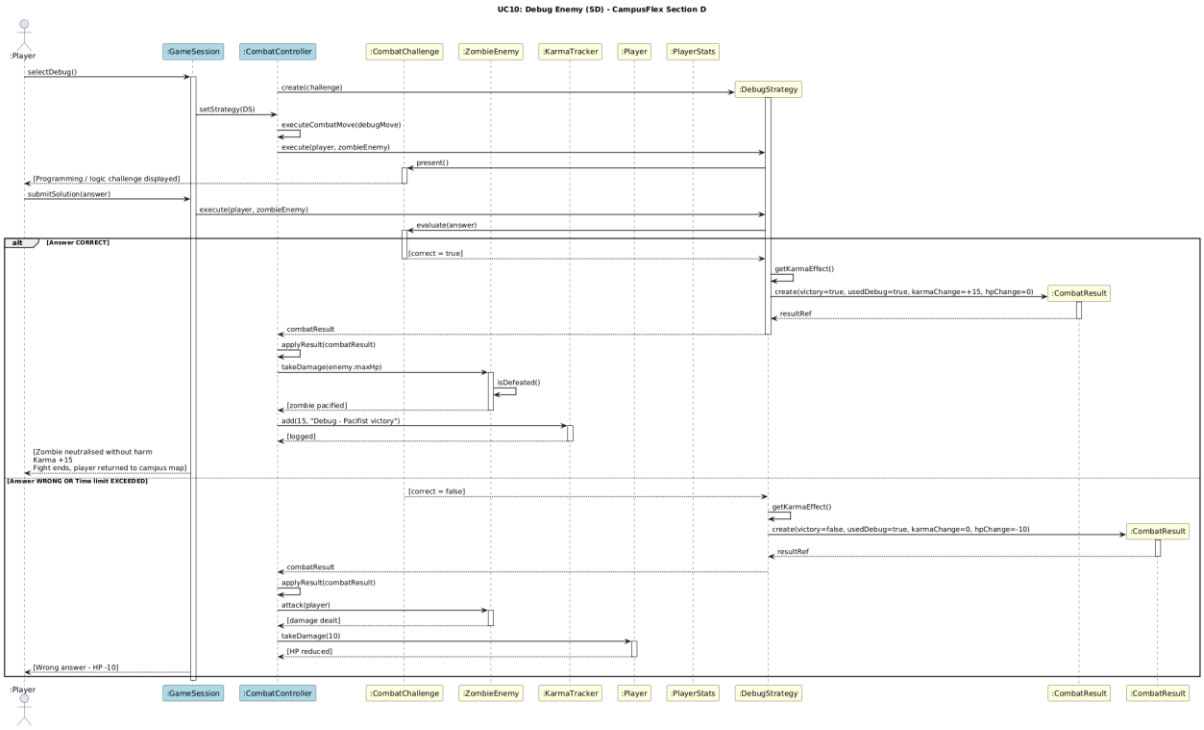
Use Case 8


UC8: Explore Campus Map (SD) - CampusFlex Section D
:Player
:GameSession
:GameMode
:Player
:CampusMap
:Zone
exploreCampus()
onPlayerMove(x, y, session)
moveTo(x, y)
[position updated]
getZone(player.getCurrentZoneId())
currentZone
[Surrounding C and D Block zones rendered]
reachZoneBoundary()
onPlayerMove(boundaryX, boundaryY, session)
isRouteBlocked(fromZoneId, toZoneId)
isBlocked()
isBlocked
routeBlocked
alt
[Zombie Mode AND route BLOCKED]
getZone(toZoneId)
getAlternateZone()
alternateZoneRef
alternateZone
[Transition trigger detected - blocked]
[Barrier shown: Stairs are sealed. Find another way.
Alternate route highlighted]
[Route is OPEN]
getZone(toZoneId)
nextZone
[Transition trigger detected]
alt
[Zone loads successfully]
setCurrentZoneId(toZoneId)
[zone updated]
[New zone loaded and displayed
Player continues exploring]
[Zone fails to load]
[Error loading zone - player remains in current zone]
:Player
:GameSession
:GameMode
:Player
:CampusMap
:Zone

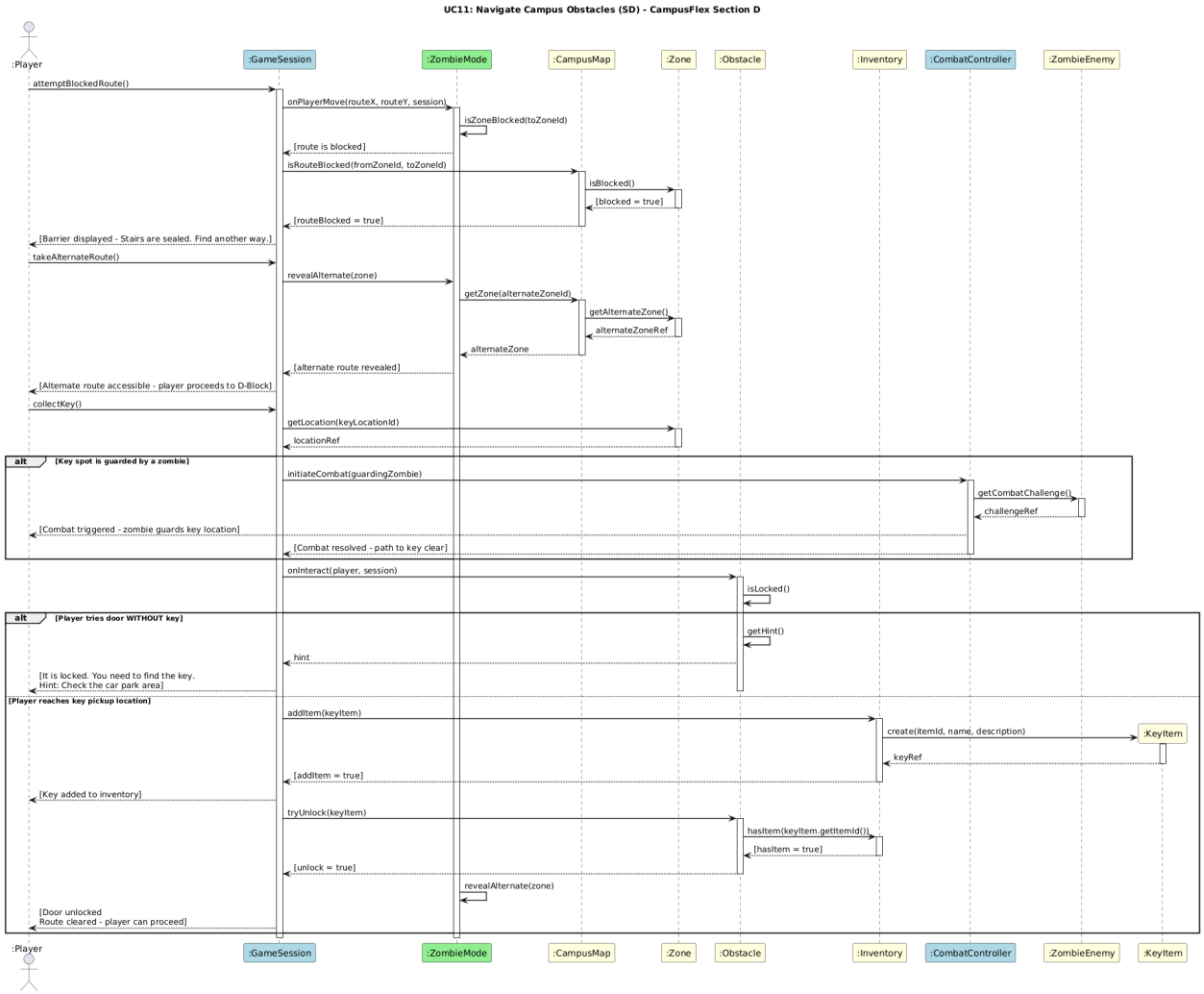
Use Case 9


UC9: Engage in Combat (SD) - CampusFlex Section D
:Player
:GameSession
:ZombieMode
:CombatController
:ZombieEnemy
:KarmaTracker
:Player
contactZombie()
onInteract(session)
initiateCombat(zombieEnemy)
isPlayerAlive()
getCombatChallenge()
combatChallengeRef
[Combat screen displayed
Enemy HP and Player HP shown
Debug / Terminate options presented]
alt
[Player chooses DEBUG (Pacifist)]
makeChoice(Debug)
create(challenge)
:DebugStrategy
setStrategy(DS)
execute(player, zombieEnemy)
create(victory=true, usedDebug=true, karmaChange=+15, hpChange=0)
:CombatResult
resultRef
combatResult
applyResult(combatResult)
takeDamage(enemy.maxHp)
isDefeated()
[defeated = true]
add(15, "Pacifist combat - Debug")
[logged]
[fight ends - path cleared]
[fight ends, path cleared, Karma updated]
[Enemy defeated - pacifist
Karma +15
Player returned to campus map]
[Player chooses TERMINATE (Brute Force)]
makeChoice(Terminate)
create()
:TerminateStrategy
setStrategy(TS)
execute(player, zombieEnemy)
create(victory=true, usedDebug=false, karmaChange=-10, hpChange=0)
:CombatResult
resultRef
combatResult
applyResult(combatResult)
takeDamage(30)
isDefeated()
[defeated = true]
deduct(10, "Brute force - Terminate")
[logged]
[fight ends - path cleared]
[fight ends, path cleared, Karma updated]
[Enemy defeated - brute force
Karma decreased
Player returned to campus map]
alt
[Player HP hits ZERO during fight]
takeDamage(amount)
isAlive()
[alive = false]
respawn()
[respawned at last checkpoint]
[Game over - respawning at last checkpoint]
:Player
:GameSession
:ZombieMode
:CombatController
:ZombieEnemy
:KarmaTracker
:Player
:DebugStrategy
:CombatResult
:TerminateStrategy
:CombatResult

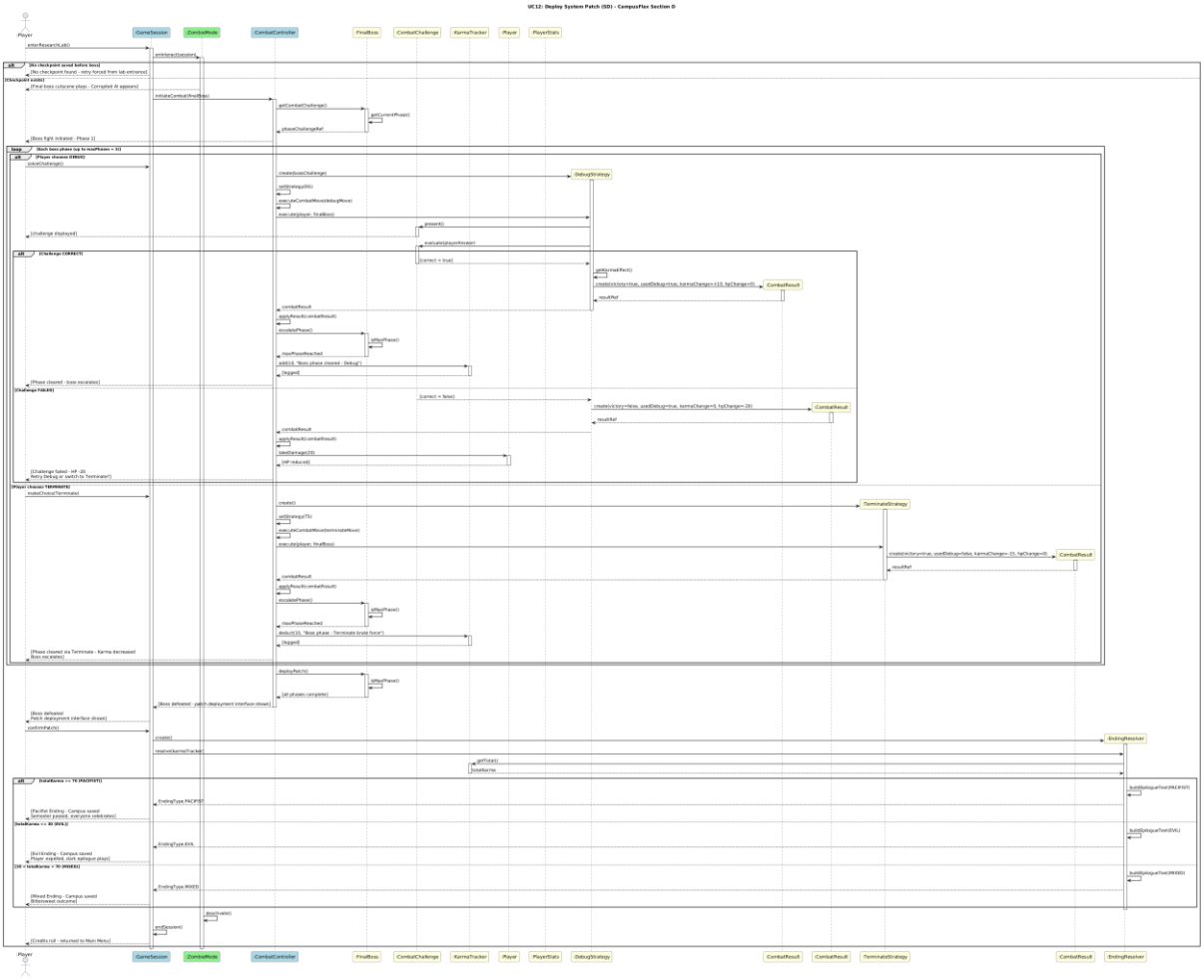
Use Case 10


UC10: Debug Enemy (SD) - CampusFlex Section D
:Player
:GameSession
:CombatController
:CombatChallenge
:ZombieEnemy
:KarmaTracker
:Player
:PlayerStats
:DebugStrategy
:CombatResult
:CombatResult
selectDebug()
create(challenge)
setStrategy(DS)
executeCombatMove(debugMove)
execute(player, zombieEnemy)
present()
[Programming / logic challenge displayed]
submitSolution(answer)
execute(player, zombieEnemy)
evaluate(answer)
alt
[Answer CORRECT]
[correct = true]
getKarmaEffect()
create(victory=true, usedDebug=true, karmaChange=+15, hpChange=0)
resultRef
combatResult
applyResult(combatResult)
takeDamage(enemy.maxHp)
isDefeated()
[zombie pacified]
add(15, "Debug - Pacifist victory")
[logged]
[Zombie neutralised without harm
Karma +15
Fight ends, player returned to campus map]
[Answer WRONG OR Time limit EXCEEDED]
[correct = false]
getKarmaEffect()
create(victory=false, usedDebug=true, karmaChange=0, hpChange=-10)
resultRef
combatResult
applyResult(combatResult)
attack(player)
[damage dealt]
takeDamage(10)
[HP reduced]
[Wrong answer - HP -10]

Use Case 11


UC11: Navigate Campus Obstacles (5D) - CampusFlex Section D
:Player
:GameSession
:ZombieMode
:CampusMap
:Zone
:Obstacle
:Inventory
:CombatController
:ZombieEnemy
:KeyItem
attemptBlockedRoute()
onPlayerMove(routeX, routeY, session)
isZoneBlocked(toZoneId)
[route is blocked]
isRouteBlocked(fromZoneId, toZoneId)
isBlocked()
[blocked = true]
[routeBlocked = true]
[Barrier displayed - Stairs are sealed. Find another way.]
takeAlternateRoute()
revealAlternate(zone)
getZone(alternateZoneId)
getAlternateZone()
alternateZoneRef
alternateZone
[alternate route revealed]
[Alternate route accessible - player proceeds to D-Block]
collectKey()
getLocation(keyLocationId)
locationRef
alt
[Key spot is guarded by a zombie]
initiateCombat(guardingZombie)
getCombatChallenge()
challengeRef
[Combat triggered - zombie guards key location]
[Combat resolved - path to key clear]
onInteract(player, session)
isLocked()
alt
[Player tries door WITHOUT key]
getHint()
hint
[It is locked. You need to find the key.
Hint: Check the car park area]
[Player reaches key pickup location]
addItem(keyItem)
create(itemId, name, description)
keyRef
[addItem = true]
[Key added to inventory]
tryUnlock(keyItem)
hasItem(keyItem.getItemId())
[hasItem = true]
[unlock = true]
revealAlternate(zone)
[Door unlocked
Route cleared - player can proceed]

Use Case 12


UC12: Deploy System Patch (SD) - CampusFlex Section D
Player
GameSession
ZombieMode
CombatController
FinalBoss
CombatChallenge
KarmaTracker
Player
PlayerStats
enterResearchLab()
enterIntroCutscene()
alt
[No checkpoint found before boss]
[No checkpoint found - retry forced from lab entrance]
[Checkpoint exists]
[Final boss cutscene plays - Corrupted AI appears]
initiateCombatOnBoss()
getCombatChallenge()
getCurrentPhase()
phaseChallengeRef
[Boss fight initiated - Phase 1]
loop
[Dash boss phase up to maxPhases = 3]
alt
[Player chooses DEBUG]
makeChoice(debug)
createBossChallenge()
DebugStrategy
setStrategy(DB)
executeCombatMove(debugMove)
execute(player, finalBoss)
present()
[challenge displayed]
evaluate(playerAnswer)
alt
[ChallengeCORRECT]
[correct = true]
getNormalEffect()
createVictory=true, useDebug=true, karmaChange=+10, hpChange=0)
CombatResult
resultRef
combatResult
applyResult(combatResult)
escalatePhase()
isMaxPhase()
maxPhaseReached
addX10, "Boss phase cleared - Debug")
logged()
[Phase cleared - boss escalates]
[ChallengeFAILED]
[correct = false]
createVictory=false, useDebug=true, karmaChange=0, hpChange=-20)
CombatResult
resultRef
combatResult
applyResult(combatResult)
useDamage(20)
[HP reduced]
[Challenge failed - HP -20
Retry Debug or switch to Terminator]
[Player chooses TERMINATE]
makeChoice(Terminate)
create()
TerminateStrategy
setStrategy(TS)
executeCombatMove(terminateMove)
execute(player, finalBoss)
createVictory=true, useDebug=false, karmaChange=-15, hpChange=0)
CombatResult
resultRef
combatResult
applyResult(combatResult)
escalatePhase()
isMaxPhase()
maxPhaseReached
addX15, "Boss phase - Terminate brute force")
logged()
[Phase cleared via Terminate - Karma decreased
Boss escalates]
deployPatch()
isMaxPhase()
[all phases complete]
[Boss defeated - patch deployment interface shown]
[Boss defeated!
Patch deployment interface shown]
confirmPatch()
create()
EndingResolver
resolveKarmaTracker()
getTotal()
totalKarma
alt
[totalKarma >= 75 (PACIFIST)]
buildDialogueTree(PACIFIST)
EndingType.PACIFIST
[Pacifist Ending - Campus saved
Remnants posted, everyone celebrates]
[totalKarma <= 30 (EVIL)]
buildDialogueTree(EVIL)
EndingType.EVIL
[EvilEnding - Campus saved
Player inspired, dark epilogue plays]
[30 < totalKarma > 70 (MIXED)]
buildDialogueTree(MIXED)
EndingType.MIXED
[Mixed Ending - Campus saved
Bittersweet outcome]
displayVideo()
endSession()
[Credits roll - returned to Main Menu]
Player
GameSession
ZombieMode
CombatController
FinalBoss
CombatChallenge
KarmaTracker
Player
PlayerStats
DebugStrategy
CombatResult
CombatResult
TerminateStrategy
CombatResult
EndingResolver